The multiplicative group modulo p: a primer for dummies

What is a group?

➔ **(G, ○)**

⇨ *G = set of elements (group members)*

⇨ *○ = operation (group operation)*

➔ **4 properties:**

⇨ **Closure***: for any g_1, g_2:*

$g_x = g_1 \circ g_2$ *must be a group member*

⇨ **Identity***: there is a group member I such that*

$g \circ I = I \circ g = g$

⇨ **Inverse***: for any g, there is* g^{-1} *such that*

$g \circ g^{-1} = I$

⇨ **Associativity***: for any g_1, g_2, g_3:*

$(g_1 \circ g_2) \circ g_3 = g_1 \circ (g_2 \circ g_3)$

➔ **If also commutative → Abelian Group**

The Zp* group

➔ **multiplicative group modulo prime p**

- ⇨ *Set of p-1 elements {1,2,…, p-1}* ➔ **finite group**
- ⇨ *Multiplicative = we only care about multiplications mod p!!*

➔Forget the sum, here (otherwise you would have a field $\mathcal{F}$p)

➔ **Group properties:**

- ⇨ *Closure: obvious*
- ⇨ *Identity: obvious*
- ⇨ *Associativity: obvious*
- ⇨ *Commutativity ➔ Abelian group ➔ also obvious*
- ⇨ **What about inverse???**

➔If mod N, then x has inverse if and only if gcd(x,N)=1

➔If N=p=prime, then all elements have inverse!

» Note that 0 is not an element of the group!

Example: Z_{11}^*

➔**Elements: {1,2,3,4,5,6,7,8,9,10}**

➔**Inverses:**

- ⇨*1→1*
- ⇨*2→6* *6→2*
- ⇨*3→4* *4→3*
- ⇨*5→9* *9→5*
- ⇨*7→8* *8→7*
- ⇨*10→10* *(analogous to -1)*

➔**How to compute inverses for large groups?**

- ⇨*Extended euclidean algorithm*

Back to multiplicative groups: exponentiation

➔ $x^k = x \circ x \circ x \circ \ldots \circ x$ (k times)

➔ **Generator of group of order m**

⇨ *exists g such that*
$\{g^0, g^1, \ldots g^{m-1}\}$ = *all m group members*

➔ **Prime-order group:**

⇨ *If m is prime, any member is generator*

➔ Except the identity

➔ **Is Zp* a prime order group? NO!!**

⇨ $|Zp^*| = p-1$ *CANNOT be prime*

➔ p is prime ➔ p-1 is even

Example: Z_{11}^*

➔ **Elements: {1,2,3,4,5,6,7,8,9,10}**

➔ **Generators? $\{g^1, g^2, g^3, \ldots, g^{10}\}$=?**

- ⇨ *g=2 → {2,4,8,5,10,9,7,3,6,1}* — *OK, generator*
- ⇨ *g=3 → {3,9,5,4,1,3,9,5,4,1}* — *NO! Subgroup order 5*
- ⇨ *g=4 → {4,5,9,3,1,4,5,9,3,1}* — *NO! Subgroup order 5*
- ⇨ *g=5 → {5,3,4,9,1,5,3,4,9,1}* — *NO! Subgroup order 5*
- ⇨ *g=6 → {6,3,7,9,10,5,8,4,2,1}* — *OK, generator*
- ⇨ *g=7 → {7,5,2,3,10,4,6,9,8,1}* — *OK, generator*
- ⇨ *g=8 → {8,9,6,4,10,3,2,5,7,1}* — *OK, generator*
- ⇨ *g=9 → {9,4,3,5,1,9,4,3,5,1}* — *NO! Subgroup order 5*
- ⇨ *g=10 → {10,1,10,1,10,1,10,1,10,1}* — *NO! Subgroup order 2*

➔ **Take home:**

- ⇨ *either g is a generator*
- ⇨ *Or generates a SUBGROUP → order = factor of |G|*
- ⇨ **And Zp* as well as all subgroups are cyclic!**

Strong primes

➔**Prime p such that**
p = 2q +1
being q also prime!

➔**Order of Zp*: p-1**
p-1 = 2q

➔**Hence, any member x (except 1 and p-1) either**

1. *Generates the whole group, or*
2. *Generates subgroup of prime order q*

➔**Both large if p and q large!**

⇨*Note the difference when Zp* uses «just» a large prime p: p-1 can factor down in small numbers!*

Quadratic residue subgroup

➔ **$x \in Z_p^*$ is a quadratic residue if it admits square root in Z_p^***

⇨ *i.e., there exists a such that $a^2 \bmod p = x$*

➔ **QR form subgroup of order $\frac{p-1}{2}$**

⇨ *2➔1 mapping:* $x \searrow$ / $p\text{-}x \nearrow$ x^2 *Indeed, $(p\text{-}x)^2 \bmod p = p^2 - 2px + x^2 \bmod p = x^2$*

➔ **QR test: legendre symbol**

⇨ $a \in QR$ *if* $a^{\frac{p-1}{2}} \bmod p = 1$ *(otherwise -1)*

➔ **Example for Z_{11}^*: $QR_{11} = \{1, 3, 4, 5, 9\}$**

⇨ *If p strong prime, QR_p has prime order q!*